

# COMP9319 Exercises

**Answers :** To be released one week later.

## Question 1

- a. Using the linear-time SA construction algorithm discussed in the class, derive the Burrows Wheeler transform  $BWT(S)$  of character sequence  $S$ :

**database\$**

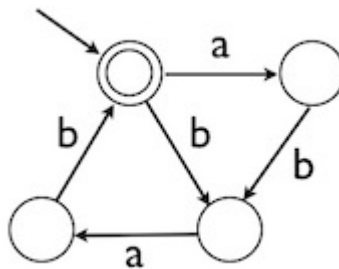
- b. Derive the Move-to-Front transform of the  $BWT(S)$  from part (a), assuming we use the 255 ASCII symbols as the symbol table.
- c. Derive and recover the original character sequence  $S$  for the  $BWT(S)$ :

**arbbr\$aa**

where  $\$$  is the pseudo end of sequence symbol.

## Question 2

- a. Derive a regular expression equivalent to the DFA:



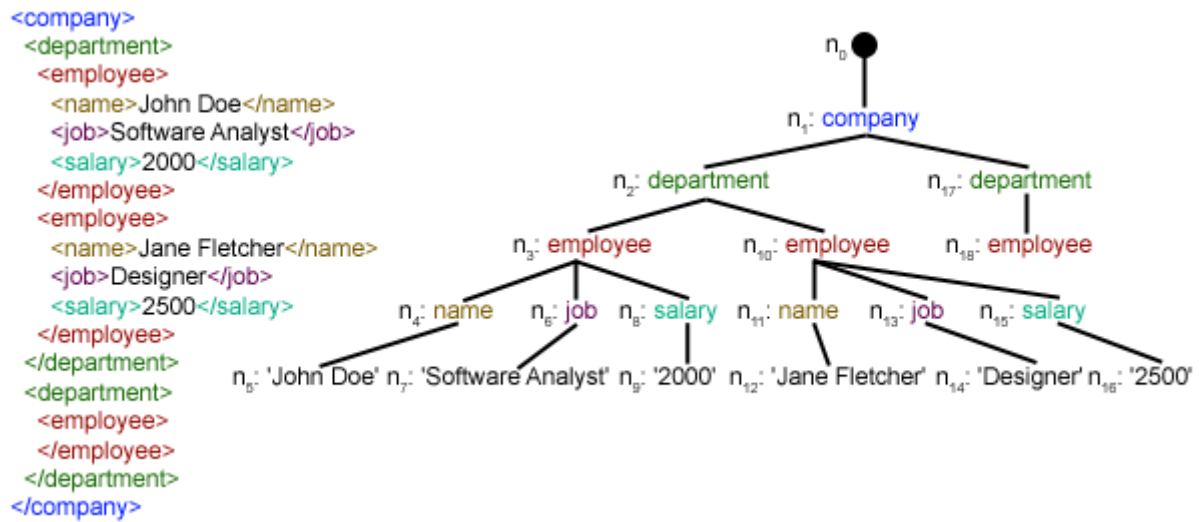
- b. Draw a DFA equivalent to the regular expression  $a^*b(alb)^*$

## Question 3

Will these two XPath expressions `/bib//book[year > 2000]/author` and `/bib[*//year > 2000]//book/author` always produce the same result on any XML documents? If not, will one of them always produce a subset of the result from the other one. Explain your answer.

## Question 4

Using the following example XML fragment and its corresponding XML tree, explain how ISX scheme can achieve efficient and yet queriable XML compression:



A Dataguide is a simple index for semistructured data. Draw the Dataguide for the XML tree above.

## Question 5

Using the example XML fragment from Question 4, explain how XBW transform works. In particular, explain how to match a sub-path in an XML tree efficiently using XBW transform.

## Question 6

Using the regular path expression **a.b+.c** and the following distributed database as an example, describe how queries on distributed semistructured data can be evaluated efficiently (with the cost of network communications minimised). Please show intermediate steps/results.

